

RESEARCH PROJECT SUMMARY

TEAM MEMBERS: 25I-7636, 25I-7601

Title: Cross-Domain Adaptation of Stacking Ensembles for Behavioral Fatigue Detection

The Concept:

The idea is to see if we can adapt a mathematical framework that is specifically designed for 'hard engineering' and apply it to 'human behavior'. We are exploring a recent study by [Farhadi et al. \(2025\)](#), which used a Stacking Ensemble to predict mechanical fatigue in structural materials. From an initial perspective, steel and the human brain have nothing in common. However, from a data perspective, both undergo a similar process of "Fatigue". Hence, the question arises, if this method is best at catching micro cracks in metal, can it also catch the micro delays in human typing patterns before a person burns out?

Problem Statement:

In the contemporary era, the majority of fatigue detection systems rely on eye-tracking cameras or EEG headsets. On the one hand, these aren't practical for the people who care about privacy or for a regular office setup. On the other hand, many developers choose deep learning models, which are black boxes and usually require massive computing power (GPUs) to run smoothly. Therefore, there is a need for a lightweight model that detects fatigue without a camera on basic CPU and uses non-invasive data, like keystroke dynamics. For this reason, we are adapting their architecture.

Proposed Methodology:

The methodology is structured into four phases, which are specifically optimized for privacy principles and CPU execution.

- **Phase 1: Data Acquisition and Pre-processing:**

We will use the [DSL-StrongPasswordData](#) dataset, which consists solely of keystroke timing latencies. In feature extraction, we will focus on Hold Time (H), Flight Time (F) and Latency (L). Then we will do labeling for fatigue, because of its session-based behavior we will apply a temporal labeling strategy.

- **Phase 2: The Level-0 "Expert" Layer:**

We will build a Heterogenous Ensemble structure by following the behavioral variance in the paper. We have selected three models such as, Random Forest, XGBoost, and SVM, to ensure diversity by looking at the data through different mathematical lenses.

- **Phase 3: The Level-1 "Meta Learner":**

In this phase we will combine the experts and implement a stacking classifier. The logistic regression acts as the fusion or the meta-learner and it learns which 'expert' is most reliable for specific typing patterns.

- **Phase 4: Performance Evaluation and Validation:**

In the final phase, the model will be evaluated using different metrics such as, accuracy, F1-Score, and AUC. Then, we will compare the stacking model's performance against each individual expert to prove the ensemble gain. Imperatively, we will document the inference time on CPU to validate that this system can run in the background of a workstation without affecting the user's computer performance.